LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	30
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Dynamic stability: longitudinal and lateral, Phugoid, Dutch roll, Fixed wing autorotation
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Dynamic stability: longitudinal and lateral, Phugoid, Dutch roll, Fixed wing autorotation. Short period oscillation, spiral divergence and Directional divergence.
5. Specific Objectives	6.1.4.1 Trainee will be able to describe aircraft stability in all the three axis.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of stability axes diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the three axes of aircraft stability?
(b) Define static stability in the context of longitudinal motion.
(c) How does dynamic stability differ from static stability?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Longitudinal dynamic stability definition • Phugoid oscillation characteristics • Short period oscillation characteristics	L & Ds	PPT, IB	15 min
(b)	• Lateral dynamic stability definition • Dutch roll oscillation characteristics • Spiral divergence and directional divergence	L & Ds	PPT, PDF	20 min
(c)	• Fixed wing autorotation concept • Stability implications of autorotation • Comparison of oscillatory modes	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between Phugoid and Short period oscillations.
(b) Describe the nature of Dutch roll motion in lateral stability.
(c) How does spiral divergence relate to directional stability?

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of dynamic stability covered in this lesson will be applied in the upcoming video session on aircraft stability (LP 31).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Video on aircraft stability	Refer to the master syllabus reference list	LP 31

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min